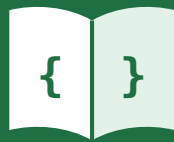

Module 03 — REST APIs with Web MVC

Exercises

Spring Boot Edu



2026-09-23

Contents

How to Work	2
Exercise 1 — Author Endpoint (Easy)	2
Exercise 2 — Order Errors as ProblemDetail (Medium)	3
Exercise 3 — Two Versions of the Order Summary (Hard)	3
Bonus Challenge (Optional)	4

How to Work

1. The starter code is in `modules/03-web-mvc/exercise/`. Find the TODO comments (Java files and `application.yaml`).
2. Every exercise already has tests, and they stay **red** until you solve it. Some tests are green from the start. They are guard tests that check your solution does not break existing behaviour.
3. Run the tests:

```
./mvnw -Pexercises -pl modules/03-web-mvc/exercise -am test
```

4. The exercise is done when all tests are green.
5. If you get stuck, read the hints first, then the solution in `modules/03-web-mvc/solution/`.

★ Tip

To run the tests of a single exercise: `-Dtest=Exercise1Test -Dsurefire.failIfNoSpecifiedTests=false`

Exercise 1 — Author Endpoint (Easy)

Goal: write a small REST endpoint with the right status codes.

Author and an `AuthorRepository` with three sample authors are provided.

Tasks (package `exercise1`):

- TODO 1a — GET `/api/authors`: all authors.
- TODO 1b — GET `/api/authors/{id}`: one author. 404 if it does not exist.
- TODO 1c — POST `/api/authors`: take a validated `AuthorRequest` and return 201 Created, a Location header and the new author.
- TODO 1d — `AuthorRequest`: the name must not be blank. The country must be exactly two upper-case letters (e.g. TR).

Hints:

- Lesson sections 3.1 and 3.2.
- The shortest way to a 404: `throw new ResponseStatusException(HttpStatus.NOT_FOUND, "...")`.
- For two upper-case letters: `@Pattern(regexp = "[A-Z]{2}")`.

Acceptance criteria: all 5 tests in `Exercise1Test` pass.

Estimated time: 30 minutes

Exercise 2 — Order Errors as ProblemDetail (Medium)

Goal: turn domain exceptions into meaningful, machine-readable problem responses.

OrderController and StockService are provided. When the stock is too low, an `OutOfStockException` is thrown, and an unknown ISBN throws an `UnknownBookException`. Right now both end in a 500 error.

Tasks (exercise2.OrderProblemHandler):

- TODO 2a — `OutOfStockException` → 409 Conflict. Type `https://springbootedu.com/problems/out-of-stock`, title `Out of stock`, detail = the exception message, instance = the request path. Extra fields: `isbn`, `requested`, `available`.
- TODO 2b — `UnknownBookException` → 404 Not Found. Type `https://springbootedu.com/problems/book-not-found`, title `Book not found`, instance = the request path, extra field `isbn`.

Hints:

- Lesson section 3.3: `ProblemDetail.forStatusAndDetail(...)`, `setType`, `setTitle`, `setInstance`, `setProperty`.
- To get the request path, add an `HttpServletRequest` parameter to the method.

Acceptance criteria: all 3 tests in `Exercise2Test` pass.

Estimated time: 30 minutes

Exercise 3 — Two Versions of the Order Summary (Hard)

Goal: serve two incompatible response formats on the same URL with Spring Framework 7 API versioning.

Today, `GET /api/order-summaries/{id}` returns a flat v1 response. The mobile app wants the currency, a readable status label and the order lines. Existing clients must not break.

Tasks:

- TODO 3a — `application.yaml`: enable versioning with the `API-Version` header.
- TODO 3b — supported versions 1 and 2; requests without the header get 1.
- TODO 3c — mark the existing method as version 1.
- TODO 3d — write the version 2 method for the same URL, and the `OrderSummaryV2` record:

```
{"id": 1001,
 "total": {"amount": 145.00, "currency": "TRY"},
 "status": {"code": "PAID", "label": "Ödendi / Paid"},
 "lines": [{"title": "Effective Java", "quantity": 1}, {"title": "Java Puzzlers",
↪  "quantity": 1}]}
```

Hints:

- Lesson section 3.5: the `spring.mvc.apiversion.*` settings and `@GetMapping(path = "...", version = "2")`.
- Nested records become nested objects in JSON: record `Money(BigDecimal amount, String currency)`.
- The status label comes from `OrderSummary.Status.label()`.

Acceptance criteria: all 4 tests in `Exercise3Test` pass.

Estimated time: 45 minutes

Bonus Challenge (Optional)

Document the author endpoint with `springdoc (@Tag, @Operation)`. Write a test that uses `@SpringBootTest(webEnvironment = RANDOM_PORT)` and `RestTestClient` instead of `@WebMvcTest`, and verify the `POST /api/authors` flow against a real server.